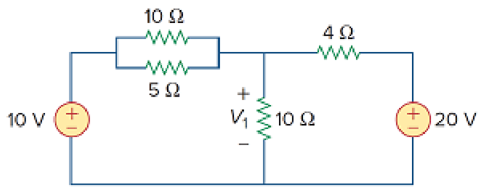
Chapter 3, Problem 6P

Problem

Solve for *V1* in the circuit of Fig. 3.55 using nodal analysis.

Figure 3.55:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.55 in the textbook.

In Figure 3.55, the resistors $10\ \Omega$ and $5\ \Omega$ are connected in parallel. Calculate the equivalent resistance.

$$\begin{aligned}10\Omega \| 5\Omega &= \frac{(10)(5)}{10+5} \\ &= \frac{50}{15}\Omega\end{aligned}$$

Step 2 of 3

Draw the modified circuit diagram.

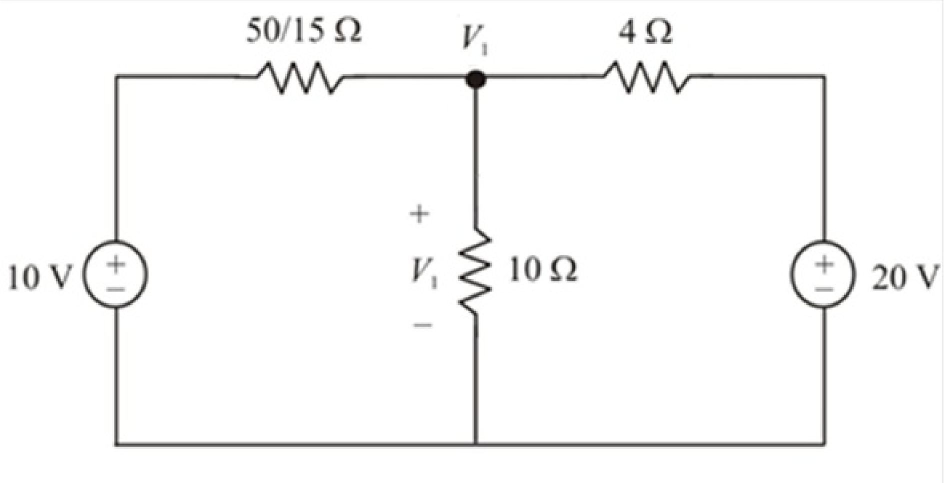


Figure 1

Step 3 of 3

Write the Kirchhoff's current law at node, V_1.

$$\frac{V_1 - 10}{\frac{50}{15}} + \frac{V_1}{10} + \frac{V_1 - 20}{4} = 0$$

$$\left(\frac{1}{\left(\frac{50}{15}\right)} + \frac{1}{10} + \frac{1}{4} \right) V_1 = \frac{10}{\left(\frac{50}{15}\right)} + \frac{20}{4}$$

$$(0.3 + 0.1 + 0.25) V_1 = 3 + 5$$

$$0.65 V_1 = 8$$

$$V_1 = 12.3 \text{ V}$$

Therefore, the voltage, V_1 is $\boxed{12.3 \text{ V}}$.